PSNA College of Engineering and Technology, Dindigul – 624 622.

(An Autonomous Institution Affiliated to Anna University, Chennai)

SECOND SERIAL TEST - Answer key

B.E Biomedical engineering Max. Marks 60

Semester III 2022

BM2312 | ELECTRONIC DEVICES & CIRCUITS

No	Part–A: Answer All Questions. 10 x 2 = 20 Marks
1	**What is the role of the small-signal model in MOSFET amplifier analysis?** The small-signal model represents a MOSFET using equivalent circuit elements around its DC operating point (Q-point), allowing the analysis of small AC variations in voltage and current. • It is used to determine voltage gain, input resistance, and output resistance. • The MOSFET is commonly represented by a dependent current source $g_m v_{gs}$ and output resistance r_o. $g_{m=\frac{\Delta ID}{\Delta VGS}}$
2	**Define the voltage gain of a common-source amplifier.** Voltage gain is the ratio of the small-signal output voltage to the small-signal input voltage. $A_{v=} V_o/V_i$ For a common-source amplifier with an unbypassed source grounded for AC and negligible channel-length modulation, $A_v = -g_m R_D$
3	**Define the common-mode rejection ratio (CMRR).** CMRR is the ratio of the magnitude of differential-mode voltage gain to the magnitude of common-mode voltage gain. $\text{CMRR} = \left\lvert \frac{A_d}{A_{cm}} \right\rvert$ In Decibels , $\text{CMRR}_{dB} = 20 \log_{10} \left\lvert \frac{A_d}{A_{cm}} \right\rvert$
4	**What is the significance of differential-mode operation?** In differential-mode operation, two input signals of equal magnitude and opposite phase are applied to a differential amplifier. $v_1 = +\frac{v_d}{2}, \qquad v_2 = -\frac{v_d}{2}$ The differential input voltage is $V_d = V_1 - V_2$ It enables amplification of the difference between two input signals while rejecting common-mode noise.
5	**Define the resonant frequency of a single-tuned amplifier?.** The resonant frequency is the frequency at which the inductive reactance equals the capacitive reactance in the parallel LC tuned circuit, producing maximum output response under ideal conditions. $X_L = X_C$, Therefore, $2\pi f_r L = \frac{1}{2\pi f_r C}$ $\boxed{f_r = \frac{1}{2\pi\sqrt{LC}}}$ where L is the inductance and C is the capacitance.

6 **Differentiate between Class A and Class B power amplifiers.**

Class A	Class B
Conduction angle is (360º).	Conduction angle is (180º) per device.
The transistor conducts throughout the input cycle.	The transistor conducts for one half-cycle.
Higher linearity but lower efficiency.	Higher efficiency but greater crossover distortion in practical push-pull circuits.
Maximum ideal efficiency is 25% for a resistively loaded series-fed circuit or 50% for a transformer-coupled circuit.	Maximum ideal efficiency is pi/4=78.5% for a sinusoid push-pull stage.

7 **What is the purpose of neutralization in high-frequency amplifiers?.**

Neutralization is a technique used to cancel the unwanted feedback caused by the transistor's internal collector-to-base capacitance C_{bc}(or equivalent reverse-transfer capacitance).

It improves amplifier stability and prevents unwanted oscillations at high frequencies.

8 **Differentiate between voltage-series and voltage-shunt feedback.**

Voltage-series feedback	Voltage-shunt feedback
Output voltage is sampled; feedback is mixed in series with the input.	Output voltage is sampled; feedback is mixed in parallel (shunt) with the input.
Increases input resistance and decreases output resistance.	Decreases input resistance and decreases output resistance

9 **What is positive feedback?.**

Positive feedback occurs when a fraction of the output signal is fed back to the input in phase with the input signal, reinforcing the input signal.

For a positive-feedback amplifier, the closed-loop voltage gain is

$$A_f = \frac{A}{1 - A\beta}$$

where A is the open-loop gain and βis the feedback factor.

Positive feedback is used in oscillators to sustain oscillations when the Barkhausen conditions are satisfied.

10 **Name the components used to determine the frequency of a Hartley oscillator.**

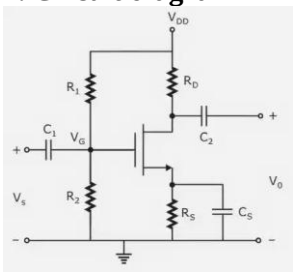
The frequency of a Hartley oscillator is determined by:

- Two inductive sections, $(L_{1)}$and$(L_{2)}$, or a tapped inductor.
- A capacitor (C) connected to form the LC resonant tank circuit.

Part–B: Answer All Questions. { (1*8) +(2*16) } = 40 Marks

11 A **Use the MOSFET small-signal model to determine the voltage gain of a common-source amplifier.**

1. Circuit diagram



2. Small-signal equivalent model

For small variations around the DC operating point, replace the MOSFET with its small-signal equivalent circuit.

- The gate draws negligible current, so input resistance is very high.
- The drain current variation is represented by the dependent current source.

$$g_m = \frac{\partial ID}{\partial VGS}$$

$$id = g_m vgs$$

If channel-length modulation is included, the output resistance is:

$$r_o = \frac{1}{\lambda ID}$$

For AC analysis, the DC supply V_D is treated as AC ground.

The source is grounded for AC, so:

$$vgs = vi$$

3. Derivation of voltage gain

The voltage gain is defined as:

$$A_v = \frac{v_o}{v_i}$$

Applying Kirchhoff's current law at the drain, neglecting r_o initially:

$$v_o = -id R_D$$

Substituting $i_d = g_m v_{gs}$:

$$v_o = -g_m v_{gsRD}$$

Since the source is grounded for AC:

$$vgs = v_i$$

Therefore,

$$A_v = \frac{v_o}{v_i} = -g_m RD$$

If the MOSFET output resistance r_o and load resistance R_L are considered, the effective drain load is:

$$RD(eff) = RD \parallel r_o \parallel RL$$

Hence,

$$A_v = -g_m(RD \parallel r_o \parallel RL)$$

If r_o is neglected and no external load is connected:

$$A_v \approx -g_m RD$$

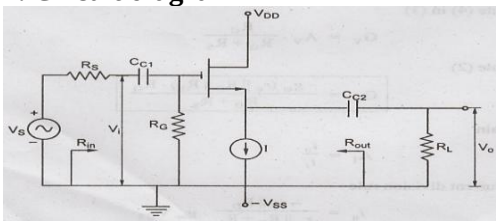
4. Interpretation and conclusion

- The negative sign indicates a phase reversal of 180° between input and output.
- Voltage gain increases with transconductance g_m and effective drain resistance.
- The amplifier has high input resistance because the gate current is approximately zero.
- It is suitable for voltage amplification in analog and sensor-interface circuits.

OR

B **Demonstrate the suitability of a MOSFET source follower for practical applications by relating its voltage gain to its output characteristics.**

1. Circuit diagram



2. Working principle and output characteristics

The input signal is applied to the gate, and the output is taken from the source.

- When the gate voltage increases, the drain current increases, raising the source voltage.
- The source voltage follows the gate voltage, with a small difference required to maintain the appropriate gate-source voltage.

$$v_o \approx v_i - VGS$$

For small-signal analysis, the DC component is removed, giving a positive, non-inverting voltage gain slightly less than unity.

3. Derivation of voltage gain

Assume the drain is at AC ground, channel-length modulation is neglected initially, and the source resistor R_S is connected to ground.

The small-signal drain current is:

$$id = g_m vgs$$

Since the output is at the source:

$$vgs = v_i - v_o$$

Applying KCL at the source:

$$\frac{v_o}{RS} = g_m(v_i - v_o)$$

Rearranging:

$$\frac{v_o}{RS} = g_m v_i - g_m v_o$$

$$v_o\left(\frac{1}{RS} + g_m\right) = g_m v_i$$

Therefore,

$$A_v = \frac{v_o}{v_i} = g_m \frac{RS}{1 + g_m RS}$$

Equivalently,

$$A_v = g_m \frac{RS}{1 + g_m RS} < 1$$

When output resistance r_o is included, replace R_S by $R_S \parallel r_o$. If a load R_L is connected, use $R_S \parallel r_o \parallel R_L$.

For $g_m R_S \gg 1$:

$$A_v \approx 1$$

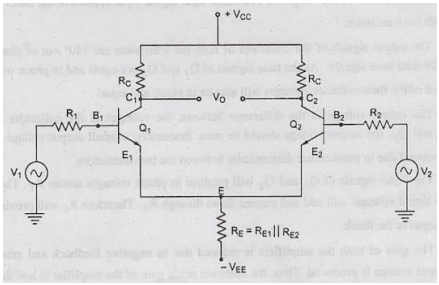
4. Practical suitability and conclusion

- High input resistance: the MOSFET gate draws negligible current.
- Low output resistance: the source follower can drive a load more effectively than a high-output-resistance voltage amplifier.
- Voltage gain close to unity, with no phase inversion.
- Used as a buffer, impedance-matching stage, and interface between high-resistance sensor circuits and lower-resistance loads.

12 A **Investigate the gain characteristics of a differential amplifier and derive the expressions for differential-mode gain and common-mode rejection ratio (CMRR).**

1. Circuit diagram and principle

A BJT differential amplifier consists of two matched transistors, Q_1 and Q_2, with their emitters connected together and supplied by a common tail-current source. The collector resistors are R,C, and the input signals are applied to the two bases.



The differential input voltage is:

$$v_d = v^1 - v^2$$

The common-mode input voltage is:

$$v_{cm} = \frac{v^1 + v^2}{2}$$

The output voltage is related to the differential and common-mode inputs by:

$$v_o = A_d\, v_d + A_{cm}\, v_{cm}$$

where A_d is the differential-mode gain and A_{cm} is the common-mode gain.

2. Differential-mode operation and gain derivation

In differential-mode operation, equal-amplitude signals of opposite phase are applied to the inputs.

$$v^1 = +\frac{v_d}{2}$$

$$v^2 = -\frac{v_d}{2}$$

For a matched differential pair with an ideal tail-current source, the common-emitter node is at AC ground under pure differential excitation.

For each transistor, the small-signal emitter-degenerated model is not needed because the common emitter node is at virtual AC ground. The small-signal collector current is:

$$i_c = g_m v_{be}$$

where:

$$g_m = \frac{I_C}{V_T}$$

Here, I_C is the quiescent collector current of each transistor and V_T is the thermal voltage, approximately 25–26 mV at room temperature.

For Q_1:

$$i_{c1} = g_{m\left(\frac{v_d}{2}\right)}$$

The corresponding collector output is:

$$v_{o1} = -i_{c1} R_C$$

$$v_{o1} = -\frac{g_m R_C v_d}{2}$$

For Q_2:

$$i_{c2} = g_{m\left(-\frac{v_d}{2}\right)}$$

$$v_{o2} = -i_{c2} R_C$$

$$v_{o2} = +\frac{g_m R_C v_d}{2}$$

Thus, for the differential output:

$$v_{od} = v_{o1} - v_{o2}$$

$$v_{od} = -\frac{g_m R_C v_d}{2} - \frac{g_m R_C v_d}{2}$$

$$v_{od} = -g_m R_C v_d$$

Therefore, the differential-output voltage gain is:

$$A_d = \frac{v_{od}}{v_d}$$

$$A_d = -g_m R_C$$

For a single-ended output taken from one collector, the gain magnitude is half of the differential-output gain:

$$A_{d(\text{single-ended})} = -\frac{g_m R_C}{2}$$

The sign depends on which input and collector output are selected. The essential result is that the differential output responds to the difference between the two inputs.

3. Common-mode operation and gain derivation

In common-mode operation, the same signal is applied to both transistor bases.

$$v^1 = v^2 = v_{cm}$$

Therefore,

$$v_d = v^1 - v^2 = 0$$

For a practical differential amplifier, assume each transistor has collector resistance R_C and the common tail-current source has a finite small-signal resistance R_E.

For a single-ended output, the common-mode gain magnitude can be derived using the small-signal emitter resistance and tail resistance.

The base voltage is:

$$v_{cm} = i_{e(r_e + 2R_E)}$$

where $r_e \approx \frac{1}{g_m}$.

Therefore,

$$i_e = \frac{v_{cm}}{r_e + 2R_E}$$

Since$i_c \approx i_e$,

$$v_o = -i_{cR_C}$$

Thus,

$$A_{cm} = \frac{v_o}{v_{cm}}$$

$$A_{cm} = -\frac{R_C}{r_e + 2R_E}$$

For $2R_E \gg r_e$:

$$A_{cm} \approx -\frac{R_C}{2R_E}$$

The common-mode gain approaches zero as the tail-current source resistance increases. For an ideal current source, $R_E \to \infty$, so $A_{cm} \to 0$.

Marking note: Accept an equivalent derivation using the common-mode half-circuit and the finite tail-source resistance.

4. Derivation of CMRR and significance

The common-mode rejection ratio is defined as the ratio of the magnitude of differential-mode gain to the magnitude of common-mode gain.

$$\text{CMRR} = \left|\frac{A_d}{A_{cm}}\right|$$

Using the single-ended gain expressions:

$$|A_d| = \frac{g_{mR_C}}{2}$$

$$|A_{cm}| = \frac{R_C}{r_e + 2R_E}$$

Therefore,

$$\text{CMRR} = \left(\frac{g_{mR_C}}{2}\right) \times \left(\frac{r_e + 2R_E}{R_C}\right)$$

Since $g_m = \frac{1}{r_e}$,

$$\text{CMRR} = 1 + 2g_{mR_E}$$

For $2g_{mR_E} \gg 1$:

$$\text{CMRR} \approx 2g_{mR_E}$$

In decibels:

$$\mathrm{CMRR(dB)} = 20\log^{10}\left|\frac{A_d}{A_{cm}}\right|$$

A high CMRR indicates that the amplifier amplifies the differential signal while rejecting common-mode interference, such as noise induced equally in both input lines.

5. Gain characteristics and conclusion

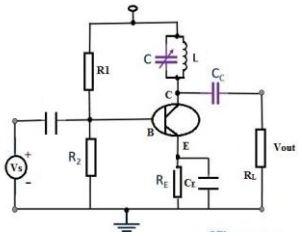
- Differential-mode gain is high when the transistors are matched and the collector loads are appropriately selected.
- Common-mode gain is low when the tail-current source has high output resistance.
- Transistor mismatch and unequal collector resistors reduce practical CMRR.
- Differential amplifiers are used in instrumentation amplifiers, operational amplifier input stages, and biomedical signal acquisition circuits.

OR

B **Assess the frequency response of a single-tuned amplifier using its resonant frequency, quality factor, and bandwidth characteristics.**

1. Definition and circuit diagram

A single-tuned amplifier is a tuned amplifier in which a single parallel LC resonant circuit is used as the collector load of a transistor amplifier. It provides maximum voltage gain at the resonant frequency and attenuates signals away from resonance.



The tuned circuit consists of inductance L and capacitance C connected in parallel. Its frequency-selective impedance is maximum at resonance, giving maximum voltage gain in the idealized tuned-load amplifier.

2. Derivation of resonant frequency

The inductive reactance is:

$$X_L = 2\pi fL$$

The capacitive reactance is:

$$X_C = \frac{1}{2\pi fC}$$

At resonance, the magnitudes of the reactances are equal:

$$X_L = X_C$$

Therefore,

$$2\pi f_{rL} = \frac{1}{2\pi f_{rC}}$$

Multiplying both sides by $2\pi f_{rC}$:

$$(2\pi f_r)^2 LC = 1$$

Hence,

$$4\pi^2 f_r^2 LC = 1$$

$$f_r^2 = \frac{1}{4\pi^2 LC}$$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

where:

- f_r = resonant frequency in hertz
- L = inductance in henries
- C = capacitance in farads

The resonant angular frequency is:

$$\omega_r = \frac{1}{\sqrt{LC}}$$

At resonance, the inductive and capacitive susceptances cancel. The tuned circuit behaves as a high impedance, producing maximum output voltage for a given AC collector current.

3. Quality factor (Q)

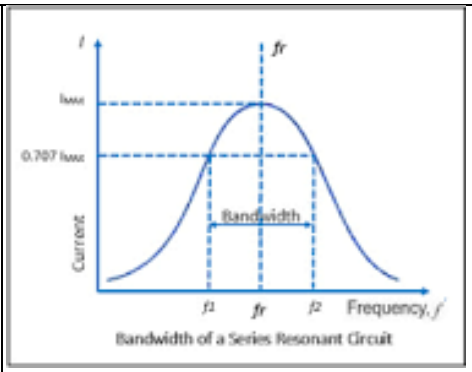
The quality factor measures the selectivity or sharpness of resonance. For a parallel RLC circuit with equivalent parallel resistance R_p:

$$Q = \frac{R_p}{\omega_{rL}}$$

Substituting $\omega_r = \frac{1}{\sqrt{LC}}$:

$$Q = \frac{R_{p\sqrt{LC}}}{L}$$

Therefore,



Bandwidth of a Series Resonant Circuit

$$Q = R_p\sqrt{\frac{C}{L}}$$

An equivalent expression is:

$$Q = \omega_{r}CR_p$$

These expressions apply to an ideal parallel RLC circuit with a parallel loss resistance R_p.

- High Q: sharper resonance, narrower bandwidth, greater selectivity.
- Low Q: broader resonance, wider bandwidth, lower selectivity.

4. Bandwidth and half-power frequencies

Bandwidth is the difference between the upper and lower half-power frequencies.

$$BW = f_H - f_L$$

At the half-power frequencies, the output power is half its maximum value. Since power is proportional to the square of voltage for a fixed resistance:

$$V_H = \frac{V_{max}}{\sqrt{2}} \approx 0.707V_{max}$$

The frequencies f_L and f_H are called the lower and upper cutoff frequencies.

For a resonant circuit with a sufficiently high Q:

$$BW = \frac{f_r}{Q}$$

Therefore,

$$Q = \frac{f_r}{BW}$$

Substituting the resonant-frequency expression:

$$BW = \frac{1}{2\pi R_{pC}}$$

for the ideal parallel RLC model.

For high-Q resonance, the approximate half-power frequencies are:

$$f_L \approx f_r - \frac{BW}{2}$$

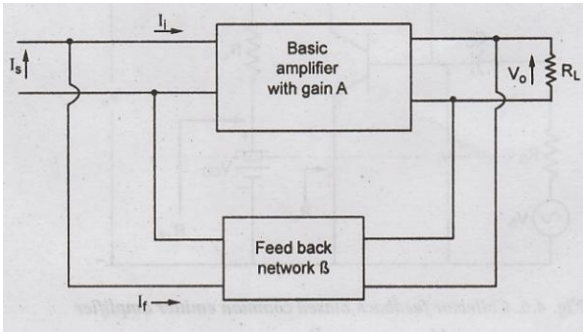
$$f_H \approx f_r + \frac{BW}{2}$$

Thus,

$$f_H - f_L \approx BW$$

13 A

Investigate the relationship between negative feedback and its effects on amplifier gain, bandwidth, stability, and distortion.



Negative feedback is the process of feeding a fraction of the output signal back to the input in such a way that it opposes the applied input signal.

The feedback signal is:

$$v_f = \beta v_o$$

The effective input is:

$$v_i' = v_i - v_f$$

where A is the open-loop voltage gain and β is the feedback factor.

2. Derivation of closed-loop gain

The output voltage is:

$$v_o = Av_i'$$

Substituting$v_i' = v_i - \beta v_o$:

$$v_o = A(v_i - \beta v_o)$$
$$v_o = Av_i - A\beta v_o$$
$$v_o(1 + A\beta) = Av_i$$

Therefore, the closed-loop gain is:

$$A_f = \frac{v_o}{v_i} = \frac{A}{1 + A\beta}$$

where:

- A = open-loop gain
- β = feedback factor
- Aβ = loop gain

For large loop gain, $A\beta \gg 1$:

$$A_f \approx \frac{A}{A\beta}$$

$$A_f \approx \frac{1}{\beta}$$

Thus, negative feedback reduces the overall gain but makes it less dependent on variations in the amplifier gain A.

3. Effect on bandwidth

For a single-pole amplifier, let the open-loop gain be:

$$A(f) = \frac{A^0}{1 + \frac{jf}{f_H}}$$

where A_0 is the low-frequency gain and f_H is the open-loop upper cutoff frequency.

The feedback gain is:

$$A_{f(f)} = \frac{A(f)}{1 + \beta A(f)}$$

Substituting:

$$A_{f(f)} = \frac{A^0}{\left[1 + A^0\beta + \frac{jf}{f_H}\right]}$$

Rearranging:

$$A_{f(f)} = \frac{\left[\frac{A^0}{1 + A^0\beta}\right]}{\left[1 + \frac{jf}{f_{H(1 + A^0\beta)}}\right]}$$

Hence:

$$A_{f0} = \frac{A^0}{1 + A^0\beta}$$

$$f_{Hf} = f_{H(1 + A^0\beta)}$$

For the same single-pole approximation, the lower cutoff frequency is reduced by the factor$(1 + A^0\beta)$. Thus, negative feedback generally increases bandwidth.

The gain-bandwidth product is approximately constant:

$$A_{f0} \times BW_f \approx A^0 \times BW$$

Key point: Gain decreases while bandwidth increases. The exact bandwidth change depends on the amplifier and feedback network.

4. Effects on stability and distortion

(i) Stability

The sensitivity of closed-loop gain to open-loop gain variation is:

$$S = \frac{\frac{\Delta A_f}{A_f}}{\frac{\Delta A}{A}}$$

$$\frac{}{f} = \frac{A}{1 + A\beta}$$

, with β constant:

$$S = \frac{1}{1 + A\beta}$$

As Aβ increases, the sensitivity decreases. Therefore, closed-loop gain becomes less affected by transistor parameter changes, temperature variation, and supply fluctuations.

Negative feedback can also improve linear stability, provided the feedback loop has adequate phase margin. Excessive phase shift can turn negative feedback into effective positive feedback at some frequencies, causing oscillation.

(ii) Distortion

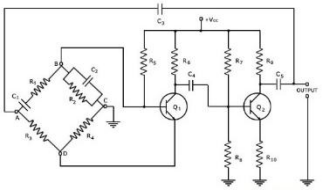
If the open-loop output contains a distortion component D, negative feedback reduces the distortion approximately to:

$$D_f = \frac{D}{1 + A\beta}$$

This is the standard approximation when the feedback reduces the distortion component by the same loop factor. Thus, negative feedback improves linearity and reduces harmonic distortion.

OR

B **Analyze the frequency-selection network and sustained oscillation conditions of a Wien bridge oscillator to determine its suitability for low-frequency sinusoidal signal generation.**



1.Definition

A Wien bridge oscillator is an RC sinusoidal oscillator that uses a lead-lag frequency-selective feedback network and a non-inverting amplifier to generate low-distortion sinusoidal signals, especially in the audio-frequency range.

The frequency-selective network consists of:

- A series RC arm.
- A parallel RC arm.
- A non-inverting amplifier.
- A negative-feedback network for setting amplifier gain and stabilizing amplitude.

2. Frequency-selection network and derivation

For the standard Wien network, the series arm contains R_1 and C_1, while the parallel arm contains R_2 and C_2.

The series-arm impedance is:

$$Z^1 = R^1 + \frac{1}{sC^1}$$

The parallel-arm impedance is:

$$Z^2 = \frac{R^2}{1 + sR^2C^2}$$

The feedback transfer function is:

$$\beta(s) = \frac{Z^2}{Z^1 + Z^2}$$

Substituting the impedances and simplifying:

$$\beta(s) = \frac{sR^2C^1}{[(1 + sR^1C^1)(1 + sR^2C^2) + sR^2C^1]}$$

Expanding the denominator:

$$\beta(s) = \frac{sR^2C^1}{[1 + s(R^1C^1 + R^2C^2 + R^2C^1) + s^2R^1R^2C^1C^2]}$$

For sinusoidal steady-state analysis, substitute $s = j\omega$:

$$\beta(j\omega) = \frac{j\omega R^2C^1}{[1 - \omega^2R^1R^2C^1C^2 + j\omega(R^1C^1 + R^2C^2 + R^2C^1)]}$$

For zero phase shift, the real part of the denominator must be zero:

$$1 - \omega^2R^1R^2C^1C^2 = 0$$

Therefore,

$$\omega^{02} = \frac{1}{R^1R^2C^1C^2}$$

Hence, the oscillation frequency is:

$$\omega^0 = \frac{1}{\sqrt{R^1R^2C^1C^2}}$$

$$f^0 = \frac{1}{2\pi\sqrt{R^1R^2C^1C^2}}$$

For equal components, $R_1 = R_2 = R$ and $C_1 = C_2 = C$:

$$f^0 = \frac{1}{2\pi RC}$$

At this frequency, the network has zero phase shift and its attenuation is:

$$\beta = \frac{1}{3}$$

for the equal-component case.

3. Barkhausen criterion and gain condition

For sustained sinusoidal oscillations, the Barkhausen criterion requires:

1. The magnitude of loop gain must equal unity.
2. The total phase shift around the loop must be 0° or an integer multiple of 360°.

Mathematically:

$$|A\beta| = 1$$
$$\angle A\beta = 0° \pmod{360°}$$

For the equal-component Wien bridge network:

$$\beta = \frac{1}{3}$$

Therefore, the amplifier gain required for sustained oscillation is:

$$A\beta = 1$$
$$A \times \frac{1}{3} = 1$$
$$A = 3$$

For a non-inverting amplifier:

$$A = 1 + \frac{R_f}{R_g}$$

Setting A = 3:

$$1 + \frac{R_f}{R_g} = 3$$
$$\frac{R_f}{R_g} = 2$$
$$R_f = 2R_g$$

Thus, the amplifier must have a gain of three at the oscillation frequency for the ideal equal-component Wien network.

4. Sustained oscillation and amplitude stabilization

At startup, the loop gain is generally made slightly greater than unity to build up oscillations from noise.

- As the output amplitude increases, an amplitude-control mechanism reduces the effective amplifier gain.
- For sustained oscillations, the loop gain settles to unity.

If the loop gain is less than unity, the oscillations decay. If it remains greater than unity, the output amplitude grows until limited by nonlinear effects, often producing distortion.

A practical Wien bridge oscillator uses a nonlinear gain-control element, such as a lamp, diodes, or an automatic gain-control circuit, to maintain stable amplitude and reduce distortion.

5. Suitability and applications

Suitability:

- Generates low-distortion sinusoidal waveforms at low and audio frequencies.
- Uses RC components instead of inductors, making it suitable for compact laboratory signal generators.
- Offers convenient frequency adjustment through R or C.
- Requires amplitude stabilization to avoid excessive distortion.

Applications: Audio-frequency generators, laboratory test instruments, amplifier testing, and low-frequency sine-wave sources.

Faculty Incharge **HOD-BME**